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Limb: Inventory-constrained design method for application of natural tree crotches as heavy timber joinery

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Abstract

LIMB is a project recently started at Taubman College at the University of Michigan to explore the potential use of natural tree bifurcations as a new joinery method in heavy timber construction.

This research studied the natural occurrence of branch bifurcations in different wood species to develop practical concepts for design and fabrication of timber joinery in a single bifurcated wood form which would replace traditional mortise and tenon or steel connections.

Through the design and fabrication of a full-scale reticulated dome, this research showed how organic variations in tree limb bifurcation brings about valuable and sustainable opportunities for generative architectural design of heavy timber construction.

Towards this goal, a hybrid physical/computational workflow was developed to study the application of these organic elements through the integration of computational design modelling and 5 axis digital fabrication. The computational workflow used the inventory of harvested tree crotches to inform the geometry of the dome to reduce material use and routing time, and moreover, the workflow evaluated the structural performance of the dome geometry using finite element analysis under self-weight and environmental loads, and provided performance and fabrication feedbacks which informed the design process.

This computational workflow was then integrated in a genetic-based form exploration process where variations in design parameters were studied through visual and quantified fabrication and structural feedback and the final geometry was chosen based on visual aesthetics and fabrication constraints.

This fabrication and inventory-aware process was a proof of concept for practical application of tree branch bifurcation for full scale construction. Beyond the reduction of waste and added value, this research explored opportunities for more organic and sustainable design and construction of heavy timber joinery in the industry.

Key words: timber joinery, tree bifurcations, wood structures, inventory-constrained design, reticulated shells, form exploration.

1_Introduction

Historic precedents for the use of tree crotches in wood structures can be found in both 17th century framing for naval vessels, as well as joinery in timber barn construction [1]. Presumably, this technique offered angular joints without complex mortise and tenon details. Additionally, this could achieve a degree of moment resistance that exceeded other joining techniques. These are demonstrated in Figure 1 which shows examples of the described use of natural tree bifurcations. In these cases, the wood grain naturally follows the shape of the piece.

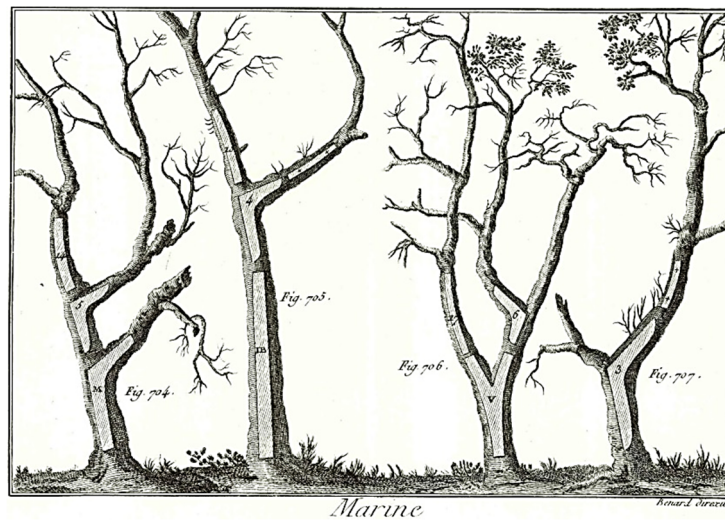


Figure 1: Bent and bifurcated elements taken from natural tree sections [1].

More recently, Claus Mattheck at the University of Karlsruhe carried out extensive physical testing of the behavior and strength of natural forked tree sections [2]. In his work he also provided guidelines for the selection and defect detection in these pieces in nature. More extensive research was carried out on structural and fracture capacity of tree bifurcations in hazel (*Corylus avellane*) by Seray Özdenlet al [3]. This research studied the toughening mechanism of the tree forks through destructive fracture test to study the fracture capacity of the interlocking wood grain that produces the critical join between branches at the apex of a hazel fork. Gareth Buckley et al [4] studied the effect of the angle of inclination on the morphology and strength of bifurcations in hazel. This can be seen in Figure 2 which shows the pattern of interlocking wood grains at the bifurcations. The bifurcations were subjected to rupture tests until they failed, and the results show that as the angle of inclination increased from 0 to 90 degrees, the rising branches became more elliptical in the vertical plane, theoretically increasing the bifurcation's bending strength in this plane. The diameter ratio was also found to be a significant factor in determining bifurcation yield strength in the samples tested.

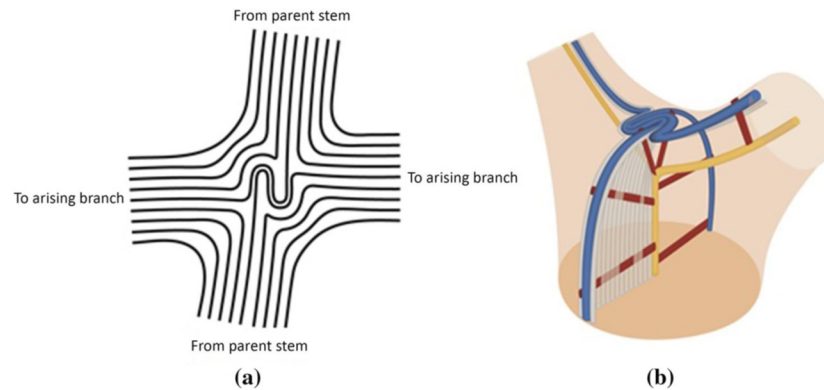


Figure 2: A Simplified pattern of the interlocking wood grain that produces the critical join between branches at the apex of a hazel fork [4]

This research studies the potentials of using tree bifurcations to take advantage of structural benefits of a joint with a single piece of wood that purposely grew under natural forces for bifurcation in nature and moving structural connection away from where vertical and horizontal members come together and develop a stronger construction joint. Beyond the reduction of waste and added value this project has the following architectural opportunities:

- Overall architectural effect relates to the natural occurrence of branch bifurcations and their formal implications in exposed structures such as roof trusses.
- A reusable system made from locally sourced materials that is environmentally conscious in its life cycle and transportable for future uses.
- The development of new ways of connecting linear timber elements beyond the “crotch” where universal scarf connections are revisited using digital fabrication.
- A structural system that is scalable to utilize many parts of a tree, from major tree bifurcations at the base suitable for large pavilions, to smaller bifurcations adaptable to furniture.

Recent researchers and designers have tried to use locally sourced trees directly in fabrication without the need for extensive processing. This included unprocessed tree bifurcations as truss elements or round timber as columns and structural elements which can be seen in Figure 3.



Figure 3_ Association School of Architecture, The Woodchip Barn, Hooke Park. WholeTrees Architecture & Structures, Festival Foods Grocery Store in Madison

This research was developed based on a syntactical approach to design. The interest was not in unique form generation based on unique parts, something that is very well explored in the realm of digital fabrication, rather are constructing a reusable language of bifurcated joinery. This research elaborates on the natural occurrence of tree bifurcation as a tectonic element using contemporary digital practices to propose new timber construction systems.

We designed a digital fabrication workflow that extracted standardized milled parts from an inventory of salvaged material. The cataloging of recurring angles and other physical properties inherent to different species of trees allowed for further development of the structural possibilities of this system, which can be applied to nearly any type of tree bifurcation. This then allows for various combinations within the language of bifurcated timber joinery seen in Figure 4. Some of these combinations include:

1. Spatially optimized parametric branching nested structure for occupiable space
2. Spatially optimized parametric hexagonal organic dome with multi angular facets
3. Three-way triangulated columnar structure
4. Two-way triangulated frame reminiscent of the traditional timber framing

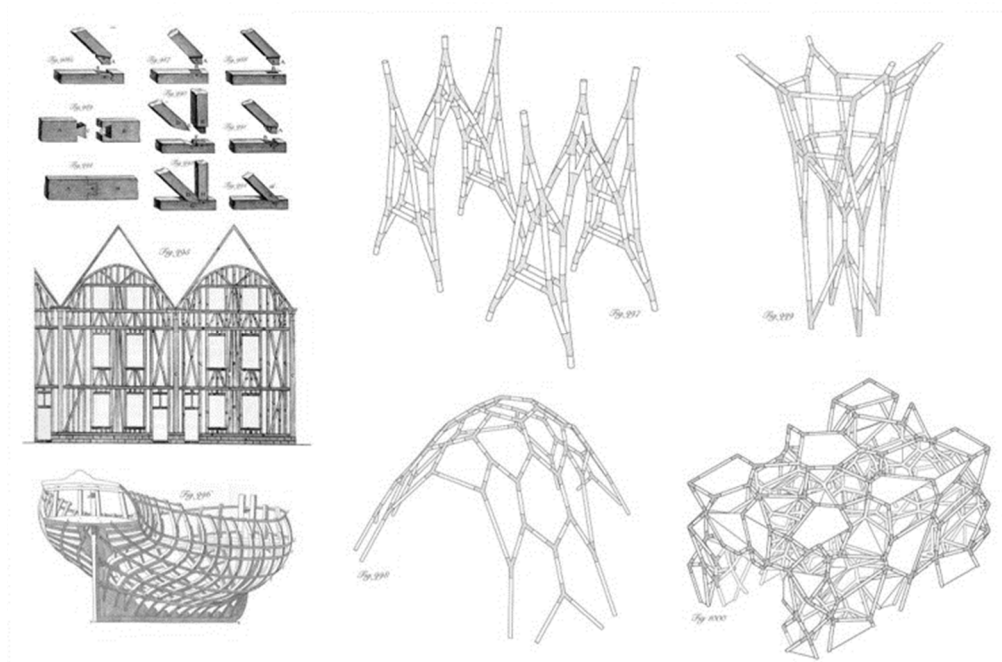


Figure 4_ different structural typologies proposed using tree bifurcation joinery.

The goal of the work presented here was to make use of these naturally forked tree sections as joining elements in architectural, structural systems. In this effort we focus on the design and fabrication of a reticulated dome structures. The raw tree crotches were cut to size and milled to final form and dimensions in a 5-axis CNC router. The design intent was to produce standardized nodal elements that can be joined easily to connective strut elements. By using the CNC router, exact final angular and linear dimensions can be precisely attained.

2_ Design process

Natural forces in trees develop adaptive growth through the process of thigmomorphogenesis, which develops organic variations in size, shape, and fiber composition of tree bifurcations. This organic variation in tree crotches requires a specific design process which accommodates the uniqueness and variation of bifurcations geometry, while also imposing a certain level of standardization for production.

Towards this goal, a computational workflow was developed to study the application of these organic elements through integration of computational design modelling and 5 axis digital fabrication. Through physical studies of the crotch geometries, key descriptive parameters of these limbs (e.g. dimensions and angular measures) were extracted as a data inventory which was then used in a computational process to determine the best fit of these crotches as structural joints.

The placement was determined through a novel two step dynamic inventory-constrained form-finding process. This process imposed the available crotch geometries into the design geometry through iterative variation of local geometry to minimize the geometric discrepancies. As a fabrication aware process in this method, global geometry responded to organic crotch geometries in a bottom-up process which first informs the location of appropriate tree crotch in the design geometry and then iteratively minimizes the discrepancies of each crotch and its corresponding joint in the design geometry leading to reduction in routing time and material use. This process transformed the design geometry to an organic shape which responds best to the available inventory of crotches. This method was implemented in Python and used Dynamic Relaxation formulation for minimization of discrepancies and is compatible with different Visual Programming Software. The process is scalable and adaptable for different design and structural typologies.

3_ Reticulated shell

Reticulated shell was the first full-scale prototype designed using the proposed method. The design process of the reticulated shell executes the following steps in the generation of design solution:

1. Selection of a base topology grid
2. Relaxation of the grid to find an optimal compression shell geometry
3. Assignment of the crotches from the inventory to the shell joints
4. Minimization the difference between the assigned crotch angles and the corresponding dome joint geometry
5. Analysis of the shell under gravity and snow loading
6. Genetic based form exploration process to find the final design solution
7. Export of performance metrics, fabrication information and images

3_1_ Selection of a base topology grid

Because tree crotches generally join three members, a hexagonal grid was chosen for the shell mesh. Also, for this initial study, it was desirable to limit the total number of joints and members to reduce fabrication time. With this specification, the grid topologies were limited in overall size and complexity. A parametric model can develop different grid topologies which control the number of joints, and some contain a central oculus that further reduce the number of joints and members. Figure 5 shows the seven topologies included in the form finding process.

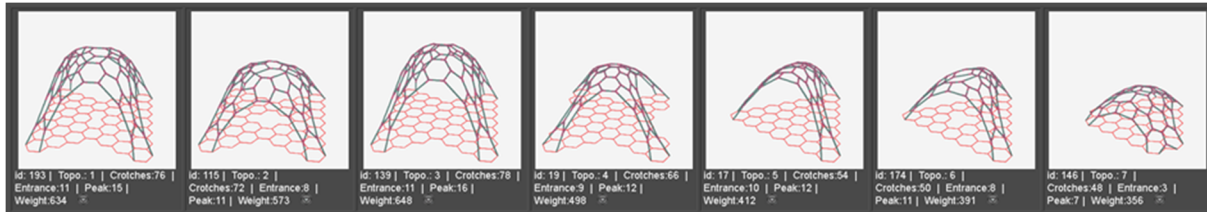


Figure 5_ Different dome topologies the parametric model can explore.

3_2_ Catenary-based for finding of the compression only dome

The next step was to use the Grasshopper plugin Kangaroo [6] to apply an upward force to find a compression-controlled shell and solve the equilibrium state using Dynamic Relaxation. The force level (rest/length ratio) which drives the height extension is taken as a variable, and spring tensions are set over three ranges of the surface – the supporting legs, the edges, and the central portion. This influences final curvature in the different regions. The examples in Figure 6 show the form-finding process.

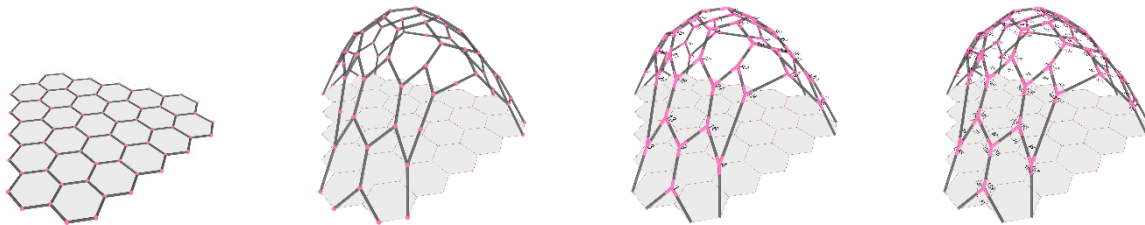


Figure 6_ Catenary-based for finding process

3_3_ Assignment of the crotches and sorting the inventory data

Once the idealistic compression-only dome geometry was generated, the digital crotches are generated at every joint of the dome and the three angles between limbs of each crotch are measured (Figure 7). In order to assign a crotch from the inventory to the closest joint in the digital model, we first had to sort the inventory data based on the angular measures. A natural way to sort the inventory was to define the controlling angles based on the morphology of each crotch. For the sake of clarity, we set up series of naming rules that determined in each tree crotch the biggest limb would be called the trunk and the two other limbs were called branches. Three angles and diameter of each limb defined the overall geometry of the crotch. The angle between branches was the smallest of the three angles and called the bifurcation angle. We sorted the inventory data based on the bifurcation angle and we set up the digital crotches vertically on their trunk limb. Then we assigned inventory items to the joints in the dome geometry based on

the smallest sum of the difference between three angles as formulated as square root of sum of square as below (Eq. 1).

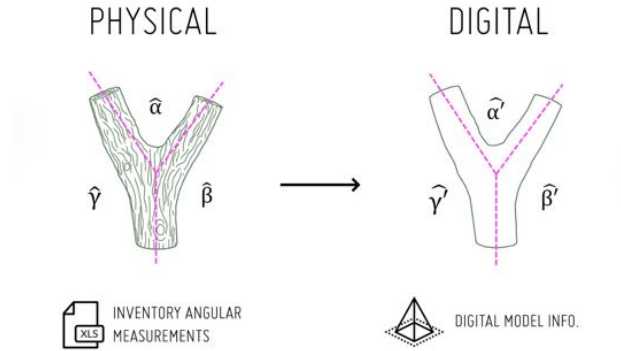


Figure 7_physical/Digital measurements

$$d_{ij} = \sqrt{(\alpha - \alpha')^2 + (\beta - \beta')^2 + (\delta - \delta')^2}$$

$$\forall i \in \{1 \dots n\}$$

$$\forall j \in \{1 \dots m\}$$

d_{ij} = SRSS of angles difference of inventory crotch
number i and digital dome joint number j
 n = number of inventory elements
 m = number of joint elements in the digital model

Equation 1_calculating angle differences to determine the closest joint for each crotch in the inventory.

3_4_Minimization of angular differences

Once all of the joints in the digital model were assigned the corresponding crotch from the inventory, we formulated the minimization process to iteratively and simultaneously minimize the discrepancies between all of the tree angles of each joint and its assigned crotch. This process was developed based on defining angular constraints between each two limb of the digital joints and constraining them within a given angular tolerance from the corresponding angle of the assigned crotch. Based on this method, the theoretical model had number of constraints equal to three times the number of joints in the model. To solve this constrained model, we had to systematically and simultaneously change the joints angles in the digital model to minimize the discrepancies. This was done using the Dynamic Relaxation method to find the overall equilibrium of the geometry by solving the coupled constrained model iteratively. This process minimally changed the local geometry of the idealistic compression-only dome to incorporate the inventory data in the model and minimize the discrepancy of the digital node and the assigned crotch geometry for each joint in the model. In effect, this process ultimately reduces the milling

time and material use, and evolves the optimal compression-only geometry to a more organic shapes which responds to the inventory (Figure 8).

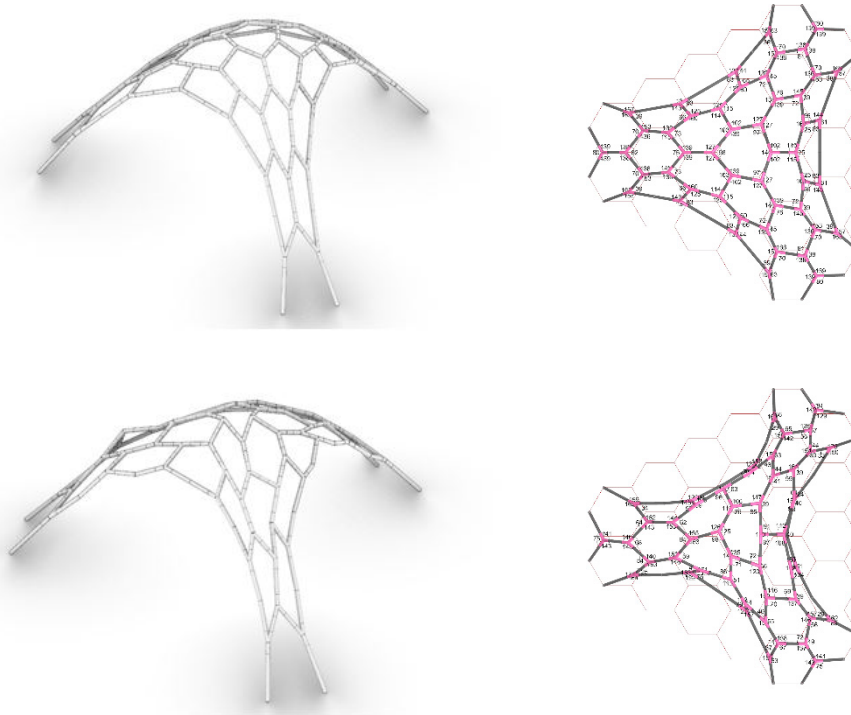


Figure 8_ Iterative process of Minimizing angular differences

3_5_ Analysis of the shell under gravity and snow load

Once the final geometry of the dome was defined this geometry was analyzed to evaluate the structural performance of the dome under self-weight and snow load. Two basic categories of data were collected for the shell: geometric parameters and structural parameters. The main geometric values included: clear height of the side arches, center maximum height, number of joints and members, longest and shortest members, defining crotch angles, and base topology type. Since the member sections and wood density were preset, the overall weight was also calculated.

The shell was considered homogeneous and the connection of members were considered fixed connections. Hardwood material properties were assigned to the members and the dome was supported using pinned supports. The dome was analyzed under self-weight and 30 psf projected snow load using linear first order structural analysis using Karamba (structural analysis plugin for Grasshopper) [7]. Values recorded from the analysis included maximum axial force, moment, and deflection, as well as utilization factors of members under axial and bi-flexural forces (Figure 9).

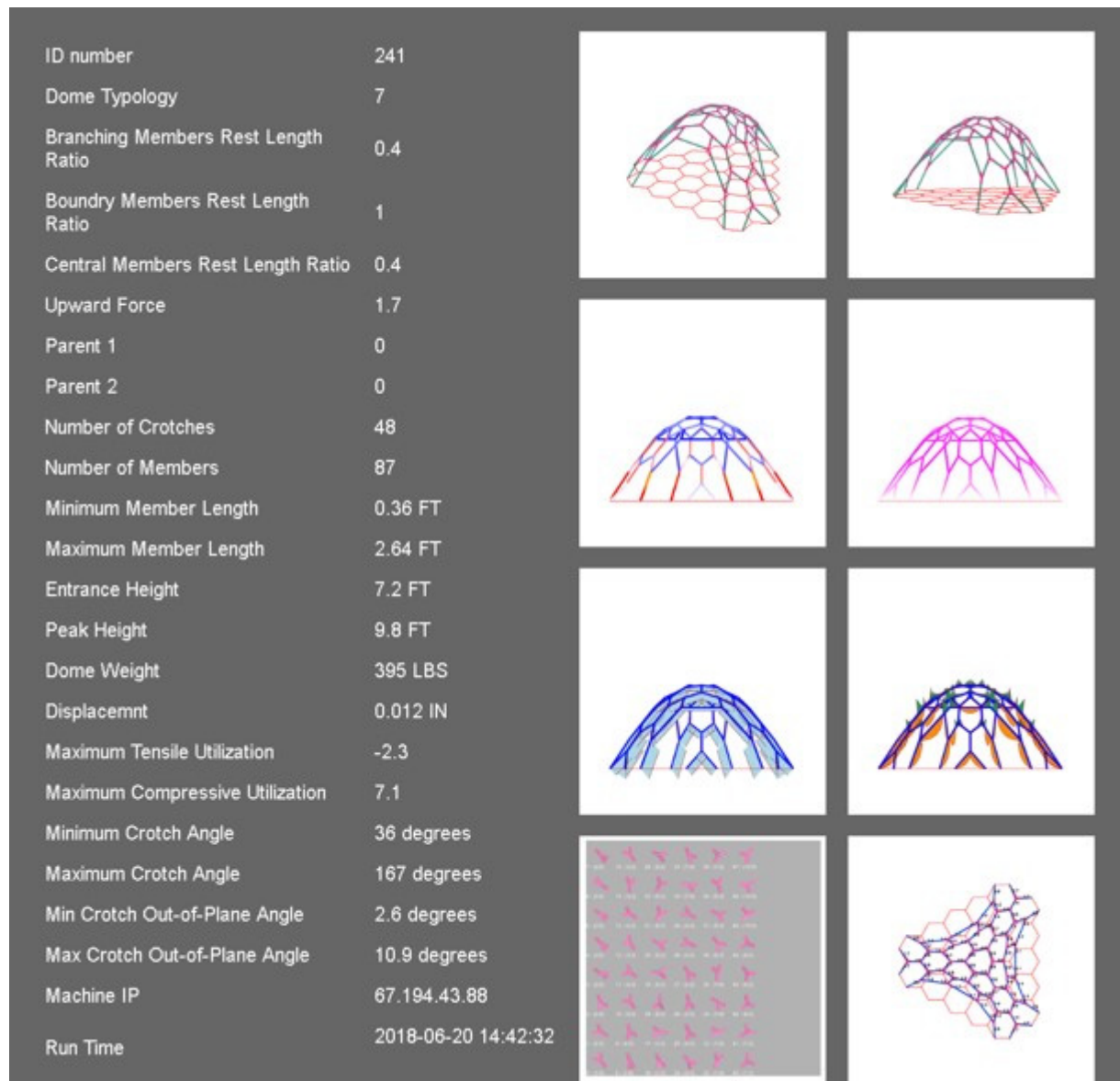


Figure 9_ Generated data for each design solution.

3_6_ Form exploration process using ParaGen

In order to find the best dome geometry based on multiple design requirements, we used a form exploration process to study and explore the design space generated by the computational model. The generation of the solution space, along with the exploration to find the best solutions was accomplished using ParaGen [8], a design aid developed at the University of Michigan. Basically, the system couples some parametric form generation with simulation and analysis tools to generate and evaluate the entire design solution. The results are uploaded to a SQL server which can then be sorted and analyzed using SQL queries. Through visual and quantitate analysis of design solutions the design space is narrowed down to find the best design solution based on visual aesthetics, performance metrics and fabrication requirements (Figure 10). Then all of the data on the final design solution are exported for fabrication process [9].

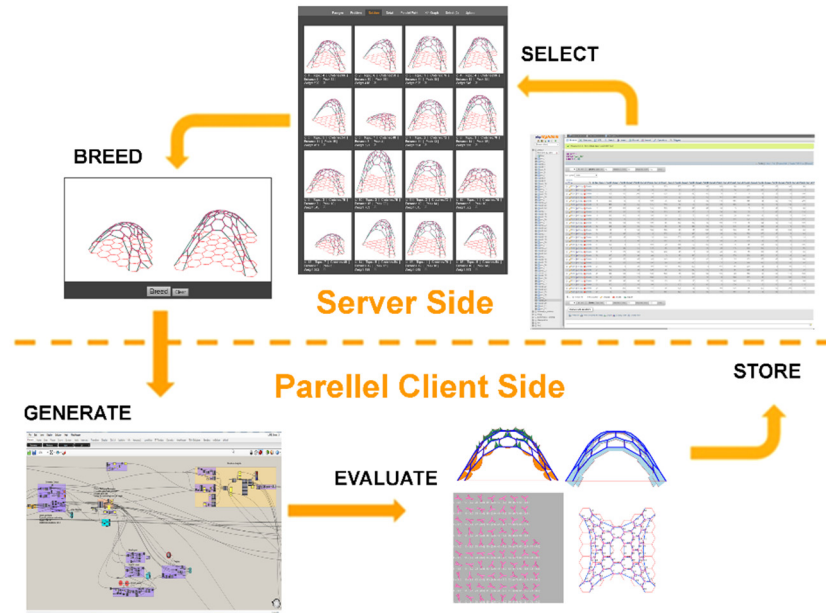


Figure 10_The ParaGen cycle showing the steps used to generate a range of solutions.

3_7_ Export of performance metrics, fabrication information and images

The computational model provided a smooth data transfer for performance metrics and fabrication data. The performance metrics and geometric information was exported into data tables and DXF files. Additionally, the final joints geometry layout was exported to then be used in the development of fabrication data, including nesting and tool pathing (Figure 11).



Figure 12_ Inventory measurements and data.

Then each crotch was scanned for using photogrammetry to generate a digital model of the crotch geometry for milling purposes.

4_3_ Milling process, physical test and post processing

Multiple 5-axis milling tests were conducted to develop an efficient fabrication process for this project. The desired finished geometry of each joinery was fit inside the corresponding scanned raw crotch geometry using 3D modeling and transformation in Rhinoceros 3D software (developed by Robert McNeel & Associates). Using Mastercam software (developed by CNC software inc.), milling toolpaths were generated for each piece and tested for the milling process (Figure 13).

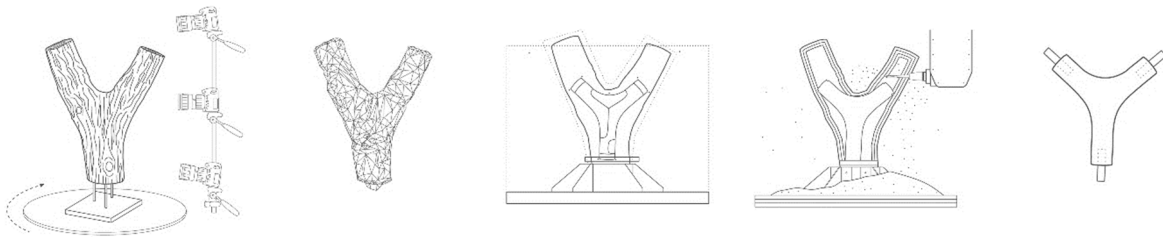


Figure 13_ Scanning and milling process.

All of the pieces were milled and sealed using a sealant to delay drying and cracking. The end of each limb of the crotch was milled to accommodate a peg connection to the linear connection member. Milled pieces were stored for assembly. Figure 14 shows a piece mounted in the milling jig at the conclusion of the 3D routing operation. Every piece went through this process which took about 2 hours each.



Figure 14_ 5-axis milling process.

4_4_ Straight member piece fabrication

The straight, strut elements between nodes were fabricated using 70 mm (2-3/4 inch) diameter billets. For these, rough turning maple blanks were used (commonly used to produce baseball bats). These are knot-free pieces, and should have a strength comparable to select mixed maple (NDS allowable compressive stress of 6000 kPa (875 psi).

4_5_ Pegged connections

The connecting element between the crotch node and the straight struts was a simple 25mm (1 inch) wood dowel used as a peg. In the original design, the intention was to glue the peg in place with casein (wood) glue (Figure 15).

The joint was tested in this form in flexure and was found to develop the full tensile strength of the dowel, which was 276 Nm (204 ft-lbs). However, because the prototype needed to be disassembled, the pegs were fixed in place with screws rather than glue, which reduced the flexure capacity to about 100 Nm (73 ft-lbs). The allowable compressive capacity would still be based on the net section and was about 20 kN (4500 lbs).

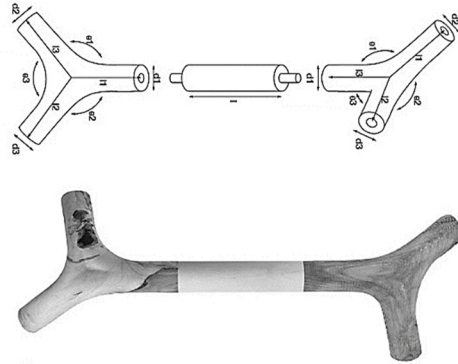


Figure 15_ Pegged connection design.

4_6_ Erection process

The erection process was easily accomplished with temporary bracing of the nodes. Due to a time limit to produce the prototype, only one third of the designed dome was fabricated and erected. Although the erection of the shell took less than one day, the production of the joints through the process described above proceeded at a rate of about two per day. In the prototype, because disassembly was necessary, the pegged joints were screwed and not glued which compromised the strength. This resulted in some joint slippage during erection (Figure 16).

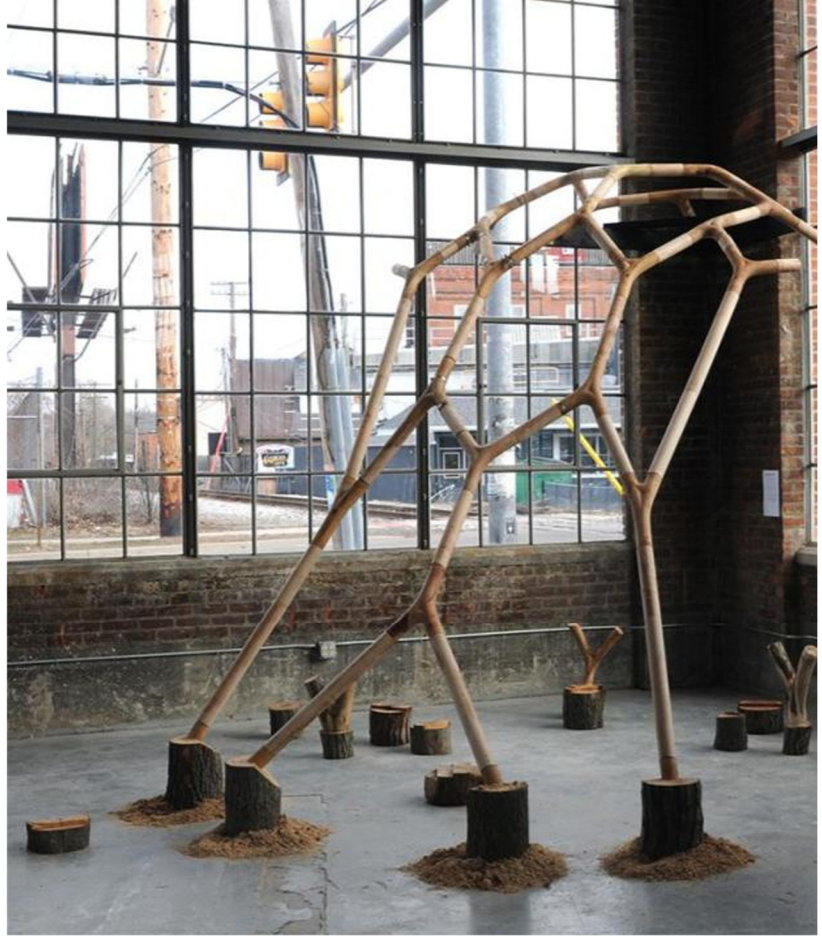




Figure 16_The constructed prototype of one third of the full dome.

5_ Conclusion and future work

The concept of repurposing the waste product of tree crotches did succeed, and a section of a dome was fabricated. The computational workflow proposed in this research effectively addressed the issues in form finding and inventory assignment, while also providing performance feedback and necessary fabrication data which is directly used for the fabrication process, this workflow can be easily adapted to respond to different structural typologies. The design exploration process with ParaGen helped the team to study and analyze the design space towards choosing the most desirable solution based on aesthetics, performance metrics, budget, and fabrication requirements.

The full-scale prototype erection process was successfully done in a day. Problems encountered include the time required to scan and mill each piece. Also, the grading of the wood was a concern. One approach would be to pre-shape pieces with a band saw, thus exposing potential defects in the wood, reducing milling time, and expediting drying. After a few weeks standing in the dry indoor environment, some cracking of the crotch pieces was also observed. As a means of crack prevention tests are currently being run using Polyethylene Glycol (PEG) injection under pressure. Also, kiln drying techniques using controlled pressure, temperature and humidity to dry the lumber without cracking are being pursued.

Acknowledgements

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